INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
SEMESTER 2, SESSION 2025/2026
SIM1006 - TUTORIAL 3

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1. Use the method discussed under 'Bernoulli equations' to solve the following problems.
   (a) $\frac{dy}{dx} - y = e^{2x}y^3$.
   (b) $3y' + y = (1-2x)y^4$.
   (c) $\frac{dx}{dy} + 3y^2x = x^2ye^{y^3}$.
   (d) $\frac{dy}{dx} + \frac{2}{x}y = x^2(x^2y)^4$

2. Classify the following equation as separable, linear, exact, or none of these. Notice that some equations may have more than one classification.
   (a) $\left(x^{\frac{10}{3}} - 2y\right)dx + x\,dy = 0$
   (b) $(ye^{xy} + 2x)dx + (xe^{xy} - 2y)dy = 0$
   (c) $y^2dx + (2xy + \cos y)dy = 0$
   (d) $\theta\,dr + (3r - \theta - 1)\,d\theta = 0$

3. For each of the following case, find the differential of the function which is in the form of $M\,dx + N\,dy = 0$. Hence, show that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
   (a) $f(x,y) = x^3y - 3y^2 = C$.
   (b) $f(x,y) = y^2e^{ax} = C$.
   (c) $f(x,y) = y^2 + \sin(2xy + x^2) = C$.

4. Show that the following differential equation is exact. Hence, solve the equations.
   (a) $[(x+1)e^x - e^y]dx - xe^ydy = 0$
   (b) $\sinh x \cos y\,dx = \cosh x \sin y\,dy$
   (c) $2\sin(2x)\sinh y\,dx = \cos(2x)\cosh y\,dy$
   (d) $2y^{-1}\cos(2x)\,dx = y^{-2}\sin(2x)\,dy$

5. Show that the following differential equation is exact. Then, solve the initial value problems.
   (a) $(y+3)\,dx + (x-2)\,dy = 0, \quad y(1) = -7$
   (b) $3x^2(y-1)^2\,dx + 2x^3(y-1)dy = 0, \quad y(-2) = \frac{5}{2}$
   (c) $\cos(\pi x)\cos(2\pi y)\,dx = 2\sin(\pi x)\sin(2\pi y)\,dy = 0, \quad y\left(\frac{3}{2}\right) = \frac{1}{2}$